

AI Agents in Trading: Where the Real Boundary of Applicability Lies

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Abstract

A practitioner-grade calibration of where LLM-based signal generation adds edge in systematic trading, where it destructively competes with classical factor methods, and what the infrastructure layer looks like when it all breaks at 9:31 AM on a CPI print day.

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Illustrative comparison (same universe · same period · different infrastructure; Jan 2019 – Dec 2024, US top-500 by ADV):

	A · Factor model	B · LLM agent
Sharpe (net)	1.0	0.3
CAGR	+14.8%	+5.2%
Max DD	−8 to −12%	−14 to −22%
IC stability	persistent	episodic
Signal character	stable infrastructure; slow alpha decay; risk priced in advance	episodic signal; alpha decays inside the news cycle; regime-fragile

Everyone is deploying AI agents in trading. Almost nobody is rigorously measuring whether they actually work — beyond the backtest, beyond the demo environment, beyond the carefully curated in-sample period that happened to include exactly the regime the model was trained on. The gap between “*our LLM agent showed 25% annualized returns in simulation*” and “*our LLM agent survived live deployment for 18 consecutive months without cascading infrastructure failures, adverse selection, or silent signal degradation*” is not a model problem. It is a systems problem — and it is precisely where most teams quietly fail, without ever publishing a post-mortem.

This article is not a defense of classical factor models. Nor is it an endorsement of the AI-agent-replaces-quant narrative that has saturated fintech discourse since late 2023. It is an attempt at a practitioner-grade calibration: where does LLM-based signal generation actually add edge, where does it destructively compete with better-established methods, and what does the infrastructure layer look like when it all breaks at 9:31 AM on a CPI print day.

1 Two paradigms, one goal: alpha generation.

The systematic equity space has operated for decades on the foundation of classical factor models. The Fama–French framework — initially three-factor, extended to five in 2015 with

the addition of profitability (RMW) and investment (CMA) factors — remains the dominant risk decomposition and return prediction paradigm for quantitative equity managers globally. Layered on top: momentum (12-1 month return, the “fourth factor” that Fama–French themselves grudgingly acknowledged), mean-reversion overlays at shorter horizons, and statistical arbitrage pairs constructed from cointegration or common-factor residuals.

The virtues of these approaches are well-documented and empirically robust across multiple decades, asset classes, and geographies: interpretability, relatively stable signal turnover, manageable transaction costs, and a research lineage dense with peer-reviewed scrutiny.^{[9][25]}

The factor model framework rests on a deceptively simple mathematical foundation. The cross-sectional expected return of asset i at time t is modelled as:

$$E[r_{i,t+1}] = \alpha_i + \sum_{k=1}^K \beta_{ik} \cdot f_{k,t} + \varepsilon_{i,t} \quad (1)$$

where $f_{k,t}$ are systematic factor exposures (value, size, profitability, investment, momentum), β_{ik} are the corresponding loadings, and α_i is the residual idiosyncratic expected return that constitutes the true alpha signal. **The entire art of quant portfolio construction lies in estimating α_i reliably enough** that, aggregated across a diversified universe of names, it generates a positive and persistent information ratio.^[28]

LLM/AI-agent signal generation is built on a fundamentally different premise: that the information content of unstructured text — earnings call transcripts, 10-K and 10-Q filings, central bank communications, analyst reports, macro press releases — is systematically underpriced by markets, and that large language models are the appropriate tool to extract that pricing discrepancy at scale.

The practical architecture varies considerably: some implementations use fine-tuned BERT-class models (FinBERT being the canonical reference) to generate cross-sectional sentiment scores as direct alpha signals; others deploy GPT-4-class models with retrieval-augmented generation pipelines to produce multi-step reasoning-based trade ideas; the most ambitious architectures build fully autonomous execution loops where an LLM agent orchestrates data ingestion, signal generation, portfolio construction, and order routing with minimal human oversight.^{[1][8][14]}

The appeal is genuine and not dismissible. FinBERT-class models can process upwards of 1,000 earnings call transcripts in the time a senior analyst team reads three. The marginal cost of expanding coverage from 500 to 5,000 names is near-zero once the pipeline is built. The risk is equally genuine: signal-to-noise ratio in NLP-derived sentiment scores is structurally lower than in factor-based signals, the turnover profile is incompatible with execution costs in most institutional contexts, and the models themselves introduce a new class of failure mode — look-ahead bias embedded in pre-training corpora — that standard backtesting frameworks are poorly equipped to detect.^{[12][55]}

2 Inputs & assumptions.

Before any empirical comparison, the experimental conditions must be stated precisely. Without this, any IC or Sharpe figure is uninterpretable:

Experiment specification	
Data	US equities, daily OHLCV plus standardized fundamental factors — EBITDA margins, book-to-market, operating profitability, asset growth / investment rate.
Period	January 2019 – December 2024. Spans two structurally different volatility regimes — the COVID shock and recovery (2020–2021) and the 2022 rate normalization cycle — as well as the 2023–2024 AI-driven equity rally.
Universe	Top 500 names by trailing 63-day average daily dollar volume. Rebalanced quarterly to maintain liquidity representativeness.
Benchmark	Equal-weighted portfolio, rebalanced monthly. Chosen over cap-weighted to avoid mega-cap concentration distorting cross-sectional signal evaluation.
LLM proxy	FinBERT sentiment score from quarterly earnings call transcripts. Normalized cross-sectionally by GICS sector. Refreshed quarterly at transcript release. ^{[7][38]}
Factor model	Fama–French 5-factor (Mkt-RF, SMB, HML, RMW, CMA) + 12-1 momentum (UMD). Exposures estimated via rolling 36-month OLS, rebalanced monthly. ^[28]
Transaction costs	Not applied in base results. Sensitivity discussed separately.

3 Empirical comparison: the numbers that matter.

Table 1: Signal property comparison: factor model vs. LLM agent (illustrative, Jan 2019 – Dec 2024)

Signal property	A · Factor model	B · LLM agent	Delta
Information Coefficient mean cross-sectional IC	0.05 – 0.07	0.02 – 0.04	danger zone
Information Ratio IC mean ÷ IC stdev	0.6 – 1.1	0.2 – 0.5	fails ICIR > 0.5
Hit rate P(correct directional call)	52 – 55%	49 – 53%	marginal vs coin-flip
Annual turnover portfolio one-way × per year	3 – 5×	8 – 15×	cost-drag eats edge
Max drawdown peak-to-trough, gross	–8% to –12%	–14% to –22%	regime fragility
Sharpe ratio gross, pre-transaction-cost	1.1 – 1.5	0.6 – 1.0	net falls to 1.0–1.1 vs 0.2–0.4

3.1 Information Coefficient.

For the Fama–French + momentum composite, an IC in the range of 0.05–0.07 is consistent with the literature: it reflects the aggregate of multiple individually weak but low-correlation sub-signals — value IC $\approx$ 0.02–0.03, momentum IC $\approx$ 0.04–0.05, profitability IC $\approx$ 0.02–0.03 — whose combination benefits from substantial diversification across signal types and holding periods.^{[9][58]}

The LLM proxy IC of 0.02–0.04 is statistically non-trivial — a mean IC of 0.03 with $N \approx 500$ passes a standard t -test — but it sits in a danger zone. At this level, even a modest increase in transaction costs can flip the strategy from modestly profitable to net-negative. The practitioner threshold — “IC below 0.03 with annual turnover above $8\times$ is almost certainly net-negative after costs” — applies directly.^[20]

There is a deeper issue: the IC of LLM sentiment signals is highly non-stationary. During earnings seasons, realized IC spikes — values of 0.06–0.08 in individual months are not unusual. Between reporting windows, IC collapses toward zero or turns negative, as the model operates on stale transcripts in a market environment that has moved on. This episodic IC structure creates a false impression of signal strength in aggregate statistics while masking the operational reality that the signal is silent for 8–10 weeks of every 13-week quarter.^{[36][62]}

Figure 01 · IC stationarity. Monthly IC, 2-year window — the signal is silent between earnings. Factor IC is roughly constant. LLM IC clusters in 4 brief windows per year. Practitioner threshold at IC = 0.03. (Illustrative — pattern from 2019–2024 backtest, two-year sample shown.)

3.2 Information Ratio.

The IR is the ratio of mean IC to its standard deviation — the Sharpe ratio of the signal itself, prior to portfolio construction. The factor model IR of 0.6–1.1 reflects both the magnitude advantage in IC and the stability advantage: factor-based signals, grounded in slow-moving fundamental revisions, generate IC that is relatively consistent across reporting windows, macro regimes, and market conditions. A standard quant PM threshold — ICIR $>$ 0.5 for signals considered for meaningful allocation — is satisfied by classical factors and failed by the LLM proxy in most parameterizations.^{[6][15]}

The LLM proxy IR of 0.2–0.5 reflects the episodic IC structure. Even if the mean IC were identical, the higher IC volatility — driven by clustering of text-based signal activity around earnings events — would substantially depress the IR. This is not a model flaw; it is a structural property of event-driven text signals operating at quarterly frequency.^[36]

3.3 Hit rate.

Factor models show hit rates of 52–55%, which is modest in absolute terms but sufficient when combined with favourable skew — correct calls tend to have larger magnitude than incorrect ones, due to the persistence of value and momentum effects. The LLM proxy hit rate of 49–53% is at best marginally better than coin-flip, and in some sub-periods (particularly the 2022 rate shock, when consensus sentiment was uniformly negative but bottom-up estimates were still being revised upward) it fell below 50%.^[13]

This reflects a known limitation of sentiment-based signals: they capture market narrative efficiently, but market narrative and subsequent returns are only loosely correlated, particularly at earnings-to-earnings horizons. A company that guided cautiously may subsequently beat revised-down estimates — a scenario where negative sentiment generates a contrarian rather than confirmatory alpha signal.

3.4 Turnover and its consequences.

The annual turnover differential — $3\text{--}5\times$ for factor models versus $8\text{--}15\times$ for LLM-based agents — is the most consequential metric in this table, and the one most frequently glossed over in academic papers.^[29]

Turnover is high in LLM-based approaches for a structural reason: the signal is document-driven. Every new transcript, every press release, every revised guidance statement has the potential to change the cross-sectional ranking of a name. Unlike a value or profitability signal — which moves only when

quarterly financials are restated, typically in a gradual and predictable direction — sentiment scores can flip sign between two consecutive documents.

At $10\times$ annual turnover on a 500-name universe with a median market cap of \$15–20 billion, a realistic one-way execution cost of 5–10 basis points translates to 50–100 basis points of annual cost drag — or 0.3–0.6 Sharpe units — which eliminates the gross edge entirely in the median parameterization.

Page-one item · any internal review of an LLM signal strategy.

*Figure 02 · Turnover $\times$ cost-drag. Net Sharpe degrades linearly with turnover. Cost regime decides where the line crosses zero. Assumes gross Sharpe = 1.2 before costs. Curves show three one-way execution cost regimes: 3 bps (best case), 5 bps (base case), 10 bps (institutional friction). Factor zone: 3–5 $\times$ turnover. LLM agent zone: 8–15 $\times$ turnover. (**Arithmetic:** net = gross – turnover $\times$ one-way cost.)*

3.5 Drawdown characteristics.

The maximum drawdown range of –14% to –22% for LLM-based agents versus –8% to –12% for factor models reflects two distinct risk dynamics. The first is regime sensitivity: sentiment models trained predominantly on post-2009 bull market transcript data are systematically underweighted toward the language patterns that appear in genuine distress scenarios. During the March 2020 COVID shock, FinBERT-derived sentiment scores provided almost no advance warning — management teams are institutionally coached to maintain confident language on calls, and crisis vocabulary only appears in extraordinary circumstances absent from the normal transcript flow.^[65]

The second dynamic is narrative crowding. When an LLM-based signal becomes widespread in the quant community — and the diffusion speed for strategies built on publicly available models like FinBERT is measured in months — the resulting factor crowding causes abrupt drawdowns when the consensus narrative shifts. Research on factor crowding dynamics shows that crowded long/short positions unwind non-linearly: the more crowded the factor, the more violent the de-crowding event, because every participant faces the same exit signal simultaneously.^{[52][54][60]}

4 Where AI agents actually win.

The empirical comparison above, taken in isolation, would suggest that LLM-based approaches are uniformly inferior. That conclusion is too strong and would be intellectually dishonest. There are specific, well-defined domains where LLM capabilities provide genuine and durable edge.

4.1 01 Alternative data parsing at scale: the earnings call anatomy.

A human analyst team of ten people cannot systematically track granular linguistic patterns across 2,000 earnings calls per quarter. FinBERT-class models can process that volume in hours, extracting features that are consistently predictive of subsequent price behavior: the density of hedging language in forward guidance sections; the sentiment divergence between prepared management remarks and analyst Q&A responses — where CFOs, off-script, tend to reveal more about actual business conditions; the frequency of clichés and vague language, which research has shown to be negatively correlated with next-quarter earnings surprise; and structural shifts in vocabulary between consecutive quarterly calls for the same company.^{[36][65][70]}

The signal from earnings call NLP is strongest not as a standalone cross-sectional ranking device, but as a **confirmation or disconfirmation signal layered on top of fundamental factor exposures**. A company that ranks in the top quintile on value and profitability factors AND shows a significant positive shift in management communication tone is a more compelling long candidate than one supported by fundamental factors alone. In this configuration — as an overlay, not a replacement — the LLM-derived signal contributes incremental IC that is partially orthogonal to classical factors, because it captures information content that does not appear in any balance sheet or income statement.^{[42][44]}

The IMF and Bank of England have both published research confirming that structured NLP analysis of corporate and central bank communications adds predictive content beyond what is captured by standard economic variables.^{[79][83]}

4.2 02 Regime detection and narrative shift identification.

This is arguably the highest-conviction LLM application in systematic investing, and it operates at a different level of abstraction than cross-sectional signal generation. Classical factor models are fundamentally cross-sectional — they rank stocks relative to each other — but they are largely agnostic about the macro regime in which that ranking is evaluated. A value factor that worked reliably from 2000 to 2006 and again from 2016 to 2022 failed categorically in 2010–2015. The factor did not change; the regime did.^{[34][56]}

The BIS published a significant working paper in 2025 — *Parsing the Pulse: Decomposing Macroeconomic Sentiment with LLMs* — demonstrating that LLM-based approaches can construct decomposed macroeconomic sentiment indices from 25 years of newspaper coverage that track hard economic data with interpretable, fine-grained decomposition across distinct sentiment dimensions: growth, inflation, financial stability, monetary policy.^{[71][73][75]} This is qualitatively different from keyword-based sentiment indices: the LLM can distinguish between “inflation rising due to supply shock” and “inflation rising due to demand overheating” — two narratives with very different implications for factor performance — from the same underlying text corpus.

For a systematic macro overlay or a factor timing model, this represents genuine edge. A regime classifier that can identify the transition from “soft landing narrative” to “stagflation narrative” in central bank and financial media text, with a lead time of 4–6 weeks relative to hard data confirmation, allows a portfolio manager to reduce value factor exposure, increase momentum exposure, and tilt toward defensive sectors — all before the consensus has repriced.^{[32][77]}

4.3 03 Multi-asset macro signal extraction from unstructured text.

Classical factor models exist primarily in the equity space. The informational connections between asset classes are often mediated through narrative — the Fed’s language shifts the rate curve, which shifts credit spreads, which shifts equity sector performance, which shifts commodity demand — and these narrative linkages are precisely what LLMs are architecturally well-suited to capture at scale.^{[33][35]}

An LLM pipeline that simultaneously ingests FOMC minutes, Treasury auction commentary, commodity supply reports from IEA and OPEC+, corporate bond market color from prime broker research, and FX positioning data can construct a cross-asset macro signal that no pure factor model can replicate, because the input data types are fundamentally heterogeneous. This is a structural advantage for multi-asset macro strategies, where the ability to synthesize across asset class narratives is the core competency. The key distinction: cross-asset macro signals derived from LLMs have a longer natural holding period than cross-sectional equity signals — macro narratives shift on timescales of months — which makes the turnover and transaction cost problem substantially less severe.

4.4 04 Autonomous research orchestration: the meta-level application.

The most durable LLM application in quant research may not involve direct signal generation at all. It operates at a meta-level: automated hypothesis generation, factor research pipeline construction, backtest code synthesis, out-of-sample validation orchestration, and — critically — systematic identification of overfitting risk and data mining bias in the research process itself.^{[19][26]}

A quant research team that uses an LLM to generate factor hypotheses from academic literature, synthesize implementation details from SSRN preprints, write initial backtesting code, flag potential look-ahead bias in data handling, run walk-forward validation, and produce a structured summary of out-of-sample degradation — that team operates at a qualitatively different research velocity than one doing all of this manually. The LLM is not generating alpha directly; it is compressing the research cycle from weeks to days, which translates into a larger set of tested ideas, more rigorous validation, and a higher

probability of finding genuine signals rather than overfitted artifacts. **The primary failure mode in quant research is not lack of ideas — it is lack of rigorous validation bandwidth.** This is the bottleneck LLMs actually address.

5 The look-ahead bias problem specific to LLMs.

There is one additional failure mode that deserves its own section, because it is specific to LLM-based approaches and has no direct analogue in classical factor model research: look-ahead bias embedded in pre-training corpora.

Standard quant backtesting assumes that the model’s inputs are point-in-time correct — that the data available at each historical decision point reflects only information actually available at that moment. This is a well-understood requirement, and most data vendors provide point-in-time databases specifically to address it.

LLMs trained on internet-scale corpora present a fundamentally different problem. A model trained on data through December 2024 has, embedded in its weights, information about which companies performed well throughout the 2019–2024 period. When this model is used to generate sentiment scores in a backtest covering the same period, the resulting performance figures are biased in ways that are extremely difficult to detect and quantify.^{[51][55][59]}

Research published at NeurIPS 2025 and on SSRN has begun to formalize look-ahead bias benchmarks for LLM-based financial models. Preliminary findings are sobering: **models evaluated without correcting for pre-training knowledge contamination show 15–30% higher gross Sharpe ratios than when evaluated on a strict out-of-sample holdout that post-dates the training cutoff.**^{[47][49][50]}

For practitioners, this means that any backtest covering a period within the model’s pre-training window is structurally compromised. The only reliable evaluation is live out-of-sample — or a holdout dataset genuinely posterior to the last pre-training data point.

6 The alpha decay acceleration problem.

Classical factor alpha decay — the documented shortening of factor holding periods as more capital pursues the same signals — has been a structural feature of systematic investing since at least 2010, when the proliferation of factor ETFs began compressing return premia associated with value, size, and momentum. The half-life of a documented factor anomaly, from academic publication to near-complete arbitrage, has shortened from approximately 10 years in the 1990s to 4–5 years by the 2020s.^{[18][20]}

*Figure 03 · Alpha decay half-life. Years from publication to near-complete arbitrage, by era. Each generation of factor research is arbitrated faster than the last. LLM-derived signals inherit and accelerate this dynamic. 1990s factor anomaly: ~10 years; 2010s factor (ETF era): 4–5 years; LLM signal (2020s): 1–2 years. (**Compression:** 10y → 4y → 1.5y; 5–7× faster over three decades.)*

LLM-based signals are subject to an accelerated version of this dynamic. FinBERT is open-source. The methodology of extracting earnings call sentiment with a fine-tuned BERT model has been described in dozens of papers, implemented in multiple commercial data products, and replicated by hundreds of quant teams globally. The alpha available to the first mover in 2019 has been substantially arbitrated away. As IBKR Research noted in a 2026 analysis, LLMs are materially shortening the shelf-life of copyable alpha — not just because more people are using the same models, but because the models themselves, by synthesizing and redistributing research, accelerate the diffusion of any insight they are trained on.^[17]

The implication for portfolio construction is direct: LLM-derived signals should be treated as **high-decay alpha sources** allocated accordingly, and retrained at higher frequency than classical factor exposures. A signal with a half-life of 6 months should not be treated with the same holding period confidence as a signal whose persistence has been documented across multiple decades.^[18]

7 The infrastructure problem nobody talks about.

Here is the failure mode that does not appear in academic papers, vendor presentations, or most conference talks: **the majority of AI agent failures in live trading environments are not model failures. They are infrastructure failures.** And the financial consequence of an infrastructure failure in an autonomous execution loop is categorically different from an infrastructure failure in a reporting system.

Industry data on production AI agent deployments shows failure rates of 70–95% in production environments, with data quality, freshness, and integration issues cited as the primary failure mode rather than model error.^{[64][67]} In financial trading specifically, the failure modes are more precise and more consequential: stale prices are not just noise — they are deterministic inputs to position sizing calculations; missing ticks cause volatility estimates to be serially miscalculated; API inconsistencies between data vendors — where the same field reports with different timestamp conventions on different endpoints — cause agents to act on a data state that never existed.^{[21][24]}

Figure 04 · Failure modes, ranked by detectability. Top row is where teams lose most before noticing. All three were observed when stress-testing Nyquist’s agent loops against real tick data.

7.1 Silent degradation · most dangerous.

Hardest to detect. Largest losses before discovery. Decision quality erodes gradually with no system-level alarm. The degradation is in signal quality, not system availability, so the orchestration layer reports green. By the time it surfaces in P&L, the drawdown is already significant.

Drivers: NLP corpus drift · regime change uncalibrated · cost-model stale · data-format breakage.

7.2 Race conditions in multi-agent architectures · hard to reproduce.

Per-agent correct. System-level violated. Signal, risk, and portfolio-construction agents share a common data layer without strict synchronization. Inconsistent reads produce position recommendations that violate risk limits at the moment of generation — even though each agent behaved correctly given its view.

Drivers: shared portfolio state under concurrent write · missing transactional boundaries.

These emerge when multiple specialized agents (signal generation, risk management, portfolio construction, order routing) share a common data layer without strict synchronization guarantees. If the signal agent and the risk agent are reading portfolio state simultaneously while the portfolio construction agent is writing to it, the resulting inconsistent reads can produce position recommendations that violate risk limits at the moment they are generated — even though each individual agent behaved correctly given its data view.^[30]

7.3 Temporal drift · most common.

Stale but accurate. Valid for yesterday, wrong for now. Agent operates on a data state that was accurate hours ago. Yesterday’s closing volatility on a day VIX has moved 15 points intraday is not a malfunction — the agent reasons correctly on wrong data, and routes orders with high confidence.

Drivers: batch SLA misses · alt-data vendor ingestion lag · microstructure data gaps · CPI / NFP / FOMC stress windows.

The most common failure mode: the agent operates on an accurate but stale data state. At the scale of minutes, this is manageable. At the scale of hours — which occurs when batch pipelines miss SLA windows, when alternative data vendors experience ingestion delays, or when market microstructure events cause data gaps — the agent’s world model diverges from market reality in ways that produce operationally valid but economically incorrect decisions. An agent that calculates a position size using yesterday’s closing volatility estimate, on a day when VIX has moved 15 points intraday, is not malfunctioning in any detectable way — it is just wrong.

7.4 Infrastructure as ground truth.

In building Nyquist’s data layer, we encountered all three of these failure modes repeatedly when stress-testing agent loops against real tick data. The most systematically damaging were events at the intersection of high message traffic and structural data complexity: opening auction periods, economic data release windows (CPI, NFP, FOMC minutes), and quarterly index rebalancing dates — events where the agent’s need for data accuracy is highest precisely when data delivery is most stressed.

Infrastructure is not a commodity layer beneath the model. For autonomous trading systems, it *is* the model’s ground truth. A model that reasons correctly on wrong data is producing wrong answers with high confidence — which is worse, from a risk management perspective, than a model that produces no answers at all.^{[21][27]}

8 Key takeaways for practitioners.

Six rules to live by:

1. **IC below 0.03 with annual turnover above $8\times$ is almost certainly net-negative after transaction costs** — this threshold should be a hard gate in any signal evaluation framework, computed before presenting gross Sharpe figures to any investment committee.
2. **LLM signals are best deployed as factor overlays, not standalone strategies** — the incremental IC contribution when added to a factor composite is more durable and more cost-efficient, because the lower net turnover of the combination reduces execution cost drag.
3. **Treat any backtest covering a period within the LLM’s pre-training window as contaminated** — performance figures are biased upward by 15–30% in gross Sharpe terms; the only reliable evaluation is live out-of-sample.^{[47][50]}
4. **Regime detection and narrative classification are the highest-conviction LLM applications** — macro narrative signals derived from central bank communication and financial media have longer natural holding periods, lower turnover profiles, and stronger structural anchors than cross-sectional earnings sentiment.^{[71][73]}
5. **Infrastructure SLAs must be defined at the agent level, not the analytics level** — data freshness requirements for autonomous execution agents are categorically stricter than for reporting workloads; point-in-time guarantees, tick completeness validation, and latency budgets under concurrent load should be contractually defined before any agent goes live.^[27]
6. **Build explicit monitoring for silent degradation** — rolling 63-day ICIR, IC trend analysis, and comparison of live IC against backtest IC distribution should be part of standard risk reporting for any LLM-based signal in production; this failure mode causes the largest losses precisely because it is undetected until the drawdown is already significant.^[47]

9 Limitations and honest caveats.

9.1 Survivorship bias.

Endemic to any study using a fixed universe defined by current liquidity. The top-500-by-volume universe as of 2024 excludes all names that were delisted, impaired, or dropped out of the investable universe during the study period — mechanically inflating backtest performance for both approaches. The effect is likely larger for LLM-based sentiment signals, which have positive coverage bias toward large, heavily covered companies with frequent earnings call content.

9.2 FinBERT as a proxy.

Represents a substantial simplification of genuine agentic behavior. A sophisticated LLM agent with tool use, retrieval-augmented generation, multi-step reasoning, and dynamic context management can produce qualitatively different — and potentially higher quality — signals than a uniform FinBERT sentiment score. The comparison presented here should be read as representative of commodity implementations, not as a ceiling on what is theoretically achievable.^{[1][76]}

9.3 Gross metrics throughout.

All Sharpe ratios, IRs, and drawdown figures are pre-transaction-cost. A factor model with 1.3 gross Sharpe and 4× turnover likely retains 1.0–1.1 net; an LLM signal with 0.8 gross Sharpe and 12× turnover likely delivers 0.2–0.4 net, if anything. This is the single most important adjustment that academic benchmarks fail to make.^[20]

9.4 Regime generalizability.

Limited. The 2019–2024 US large-cap equity universe represents a specific confluence of liquidity conditions, macro regime variety, and market structure features. The same models applied to emerging market equities, small-cap names, credit instruments, or illiquid alternatives would face different microstructure dynamics, different information availability, and different competitive landscapes. Any extrapolation requires independent empirical validation.

10 Closing.

The relevant question for quant practitioners in 2026 is not whether to use LLMs in trading — they are already being used, by everyone, whether or not teams have explicitly chosen to deploy them. The question is whether the deployment is disciplined:

- Does the team measure live IC?
- Does it account for look-ahead bias in the pre-training window?
- Does it stress-test the infrastructure layer against microstructure stress events?
- Does it have an explicit alpha decay monitoring framework?

The boundary of LLM applicability in trading is real, empirically measurable, and not where the vendor demos suggest it is. Cross-sectional equity signal generation on liquid names using public models on publicly available text is a highly competitive domain where the gross alpha is thin, the cost structure is punishing, and the decay cycle is fast. Alternative data parsing, regime classification, macro narrative synthesis, and research orchestration are the domains where LLM capabilities are genuinely differentiated — and where the infrastructure requirements, though still demanding, are compatible with the signal’s natural holding period.

The practitioners who will extract durable edge from this technology are not the ones who deployed fastest. They are the ones who measured most rigorously.

What layer are you using LLMs at in your pipeline — and what does your live ICIR decomposition actually show? We would be genuinely interested to hear from teams running these experiments in production.

If you’re running similar comparisons or hitting infrastructure bottlenecks in your agent pipelines — [let’s talk](#).

References

Numbered citations [N] map to the author’s working bibliography assembled during research — spanning Fama–French foundational papers, IMF / Bank of England / BIS working papers on text-based macro sentiment, NeurIPS 2025 look-ahead bias benchmarks, IBKR Research notes on alpha decay, and industry data on production AI agent deployment failure rates. Full source list with DOIs and direct links available on request — contact@nyquist.pro.